PHISHGUARD

Phishguard is a web-based system designed to detect phishing URLS and improve cybersecurity awareness among users.

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PROBLEM STATEMENT

In today's digital world, many users are unable to identify fake websites and malicious phishing links. Cybercriminals use these methods to steal sensitive information such as passwords, banking details, and personal data, leading to serious financial and personal loss. Existing phishing detection tools are often too technical and difficult for normal users to understand. Because of this, there is a strong need for a simple system that can help users easily detect unsafe websites and protect themselves online.

OBJECTIVES

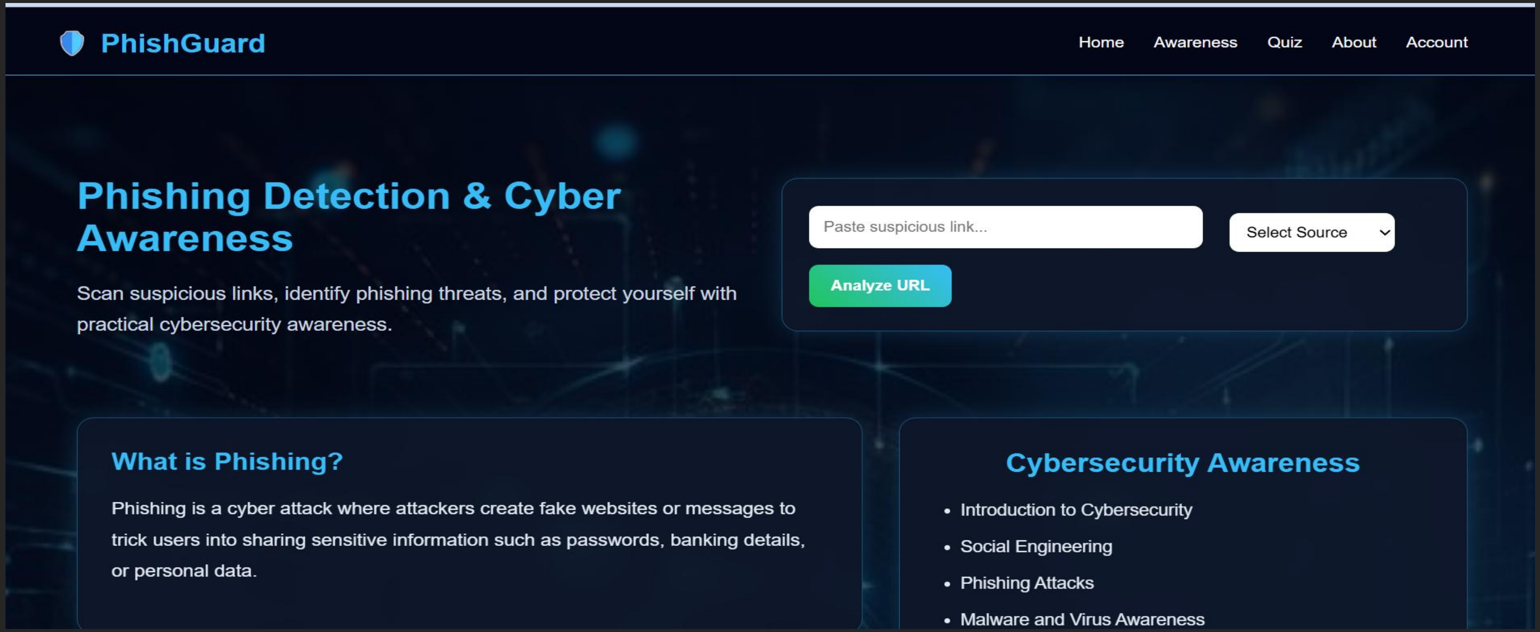
The main objective of this system is to detect phishing URLs and help users stay safe while browsing the internet. It aims to provide a simple way for users to check whether a website is safe or malicious. The system also focuses on improving cybersecurity awareness by educating users about phishing attacks and safe browsing practices.

PROPOSED SYSTEM

The proposed system allows users to enter a URL, which is then analyzed using predefined security rules. The system checks for suspicious patterns such as unsafe domains and phishing indicators. After analysis, a risk score is generated to show the safety level of the website. The result is displayed clearly, and a PDF report is generated for user reference.

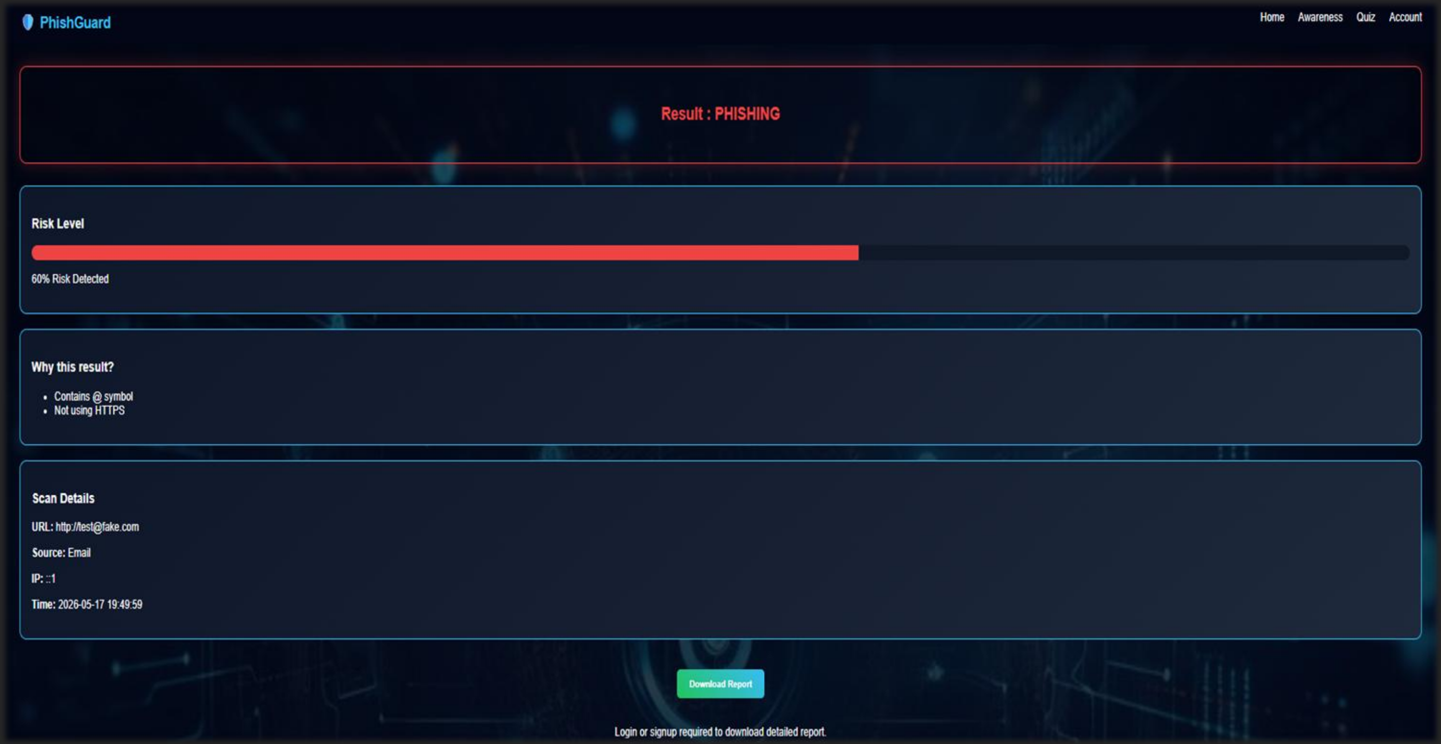
FEATURES

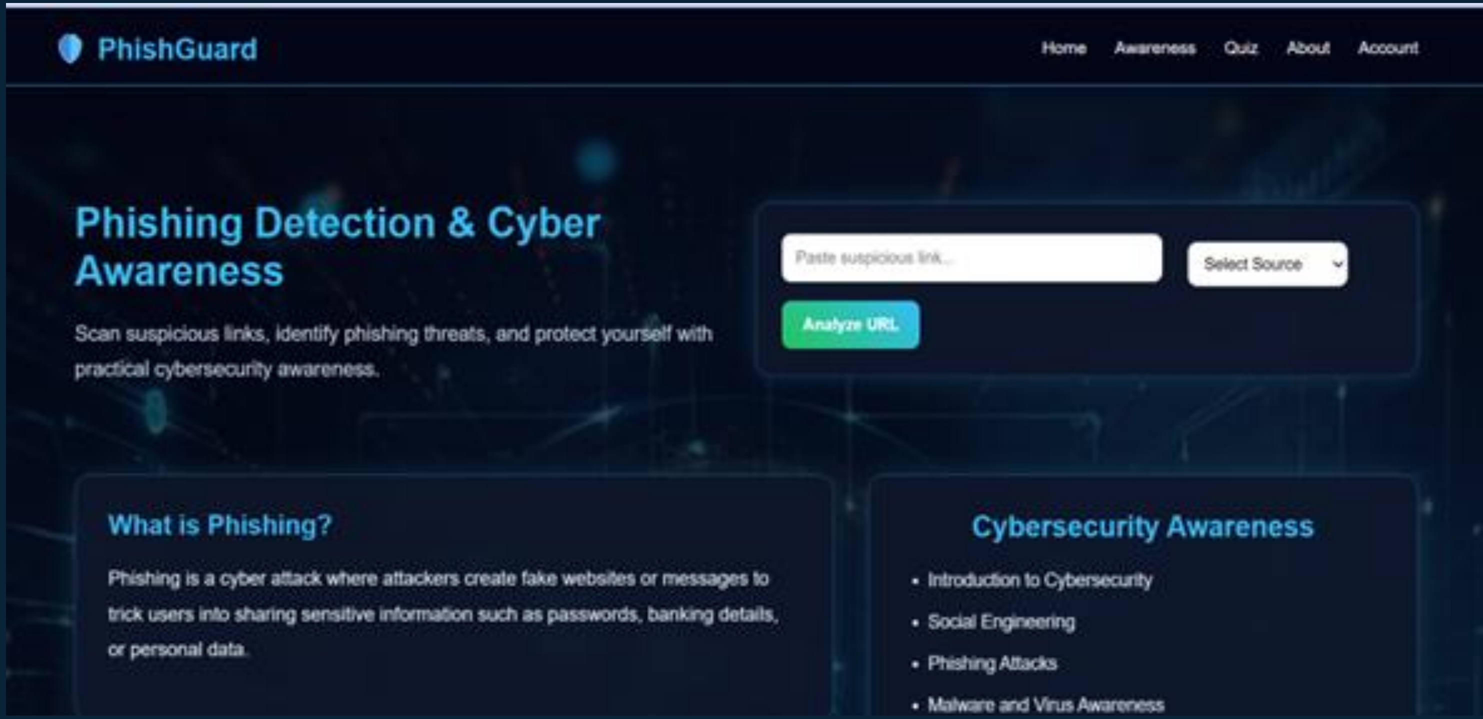
The system provides both detection and awareness features in a single platform. It helps users analyze URLs and also improves their knowledge of cybersecurity through interactive learning. It includes features like phishing detection, awareness content, quiz modules, dashboard history, and PDF report generation for better user experience.



URL Uploading page

URL Result page





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TECHNOLOGIES USED

The system is developed using modern web technologies to ensure smooth performance and easy accessibility. The frontend is designed using HTML, CSS, and JavaScript for user interaction. The backend is implemented using PHP and Python, while MySQL is used for database management. XAMPP is used as the local server environment for testing and execution.

SYSTEM ARCHITECTURE

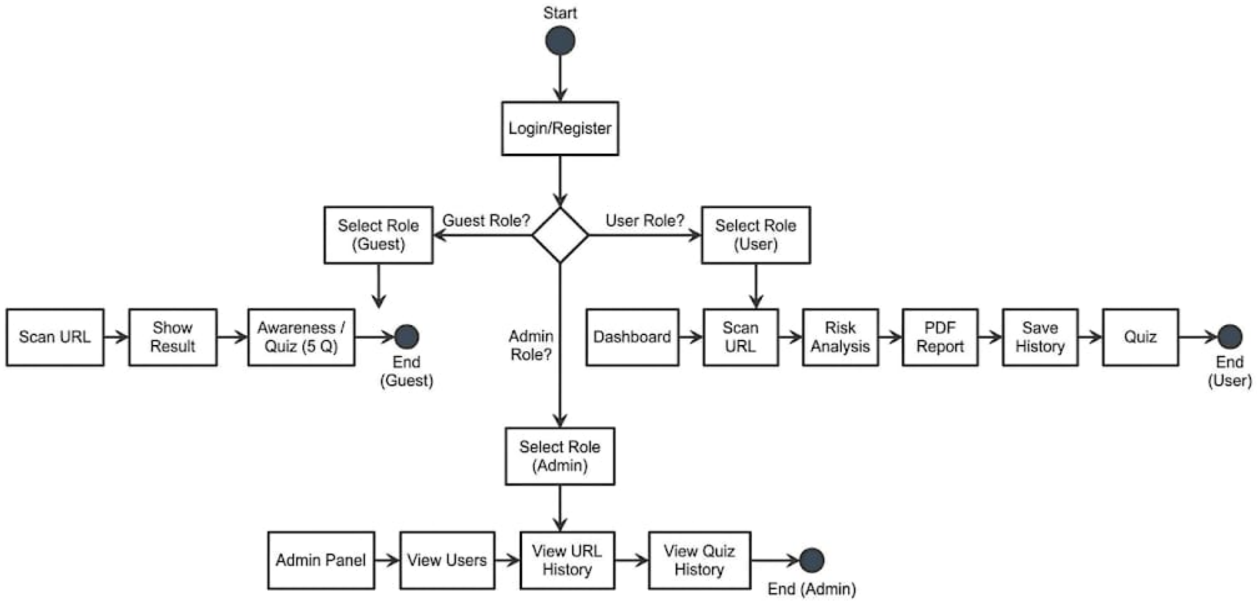
The system architecture consists of multiple layers that work together to process user requests. The frontend handles user interaction, while the backend processes the URL and performs analysis. The database stores user data and scan history, and the analysis engine evaluates URLs to detect phishing patterns and generate results.

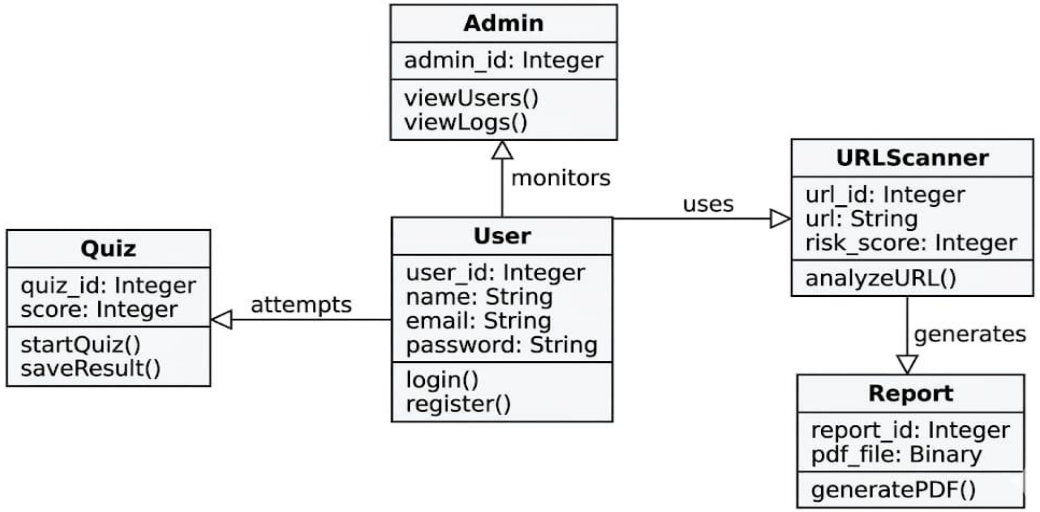
UML DIAGRAMS

UML diagrams are used to represent the structure and behavior of the PhishGuard system. They help in understanding how different components of the system interact with each other. These include Use Case, Class, Activity, and Sequence diagrams that explain system functionality clearly.

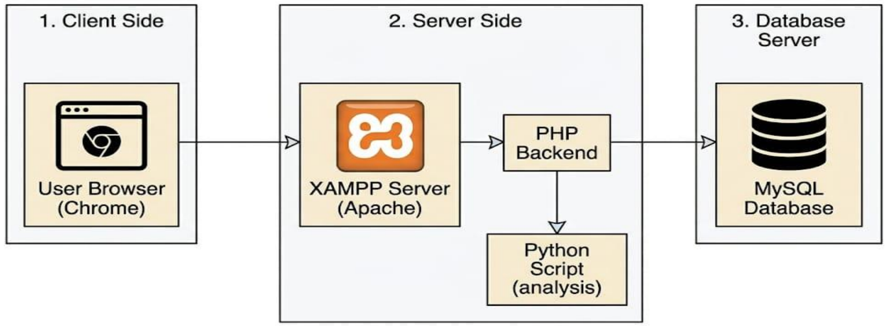
UML DIAGRAMS

PhishGuard Activity Diagram

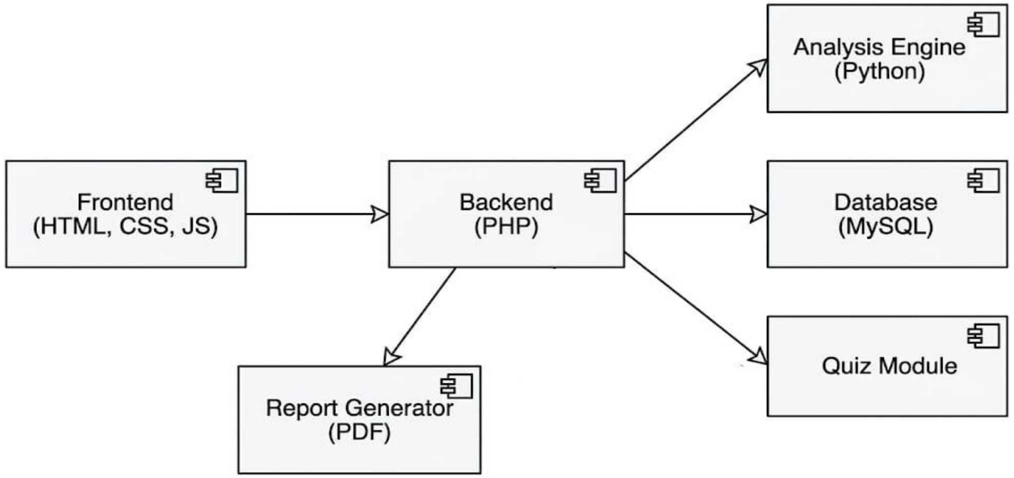




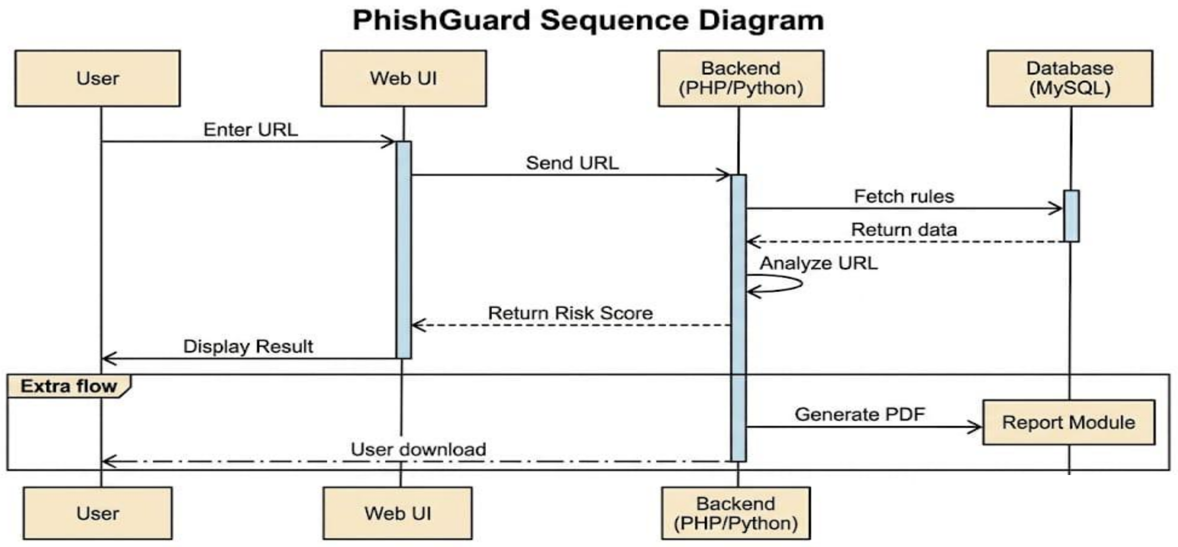
PhishGuard Deployment Diagram

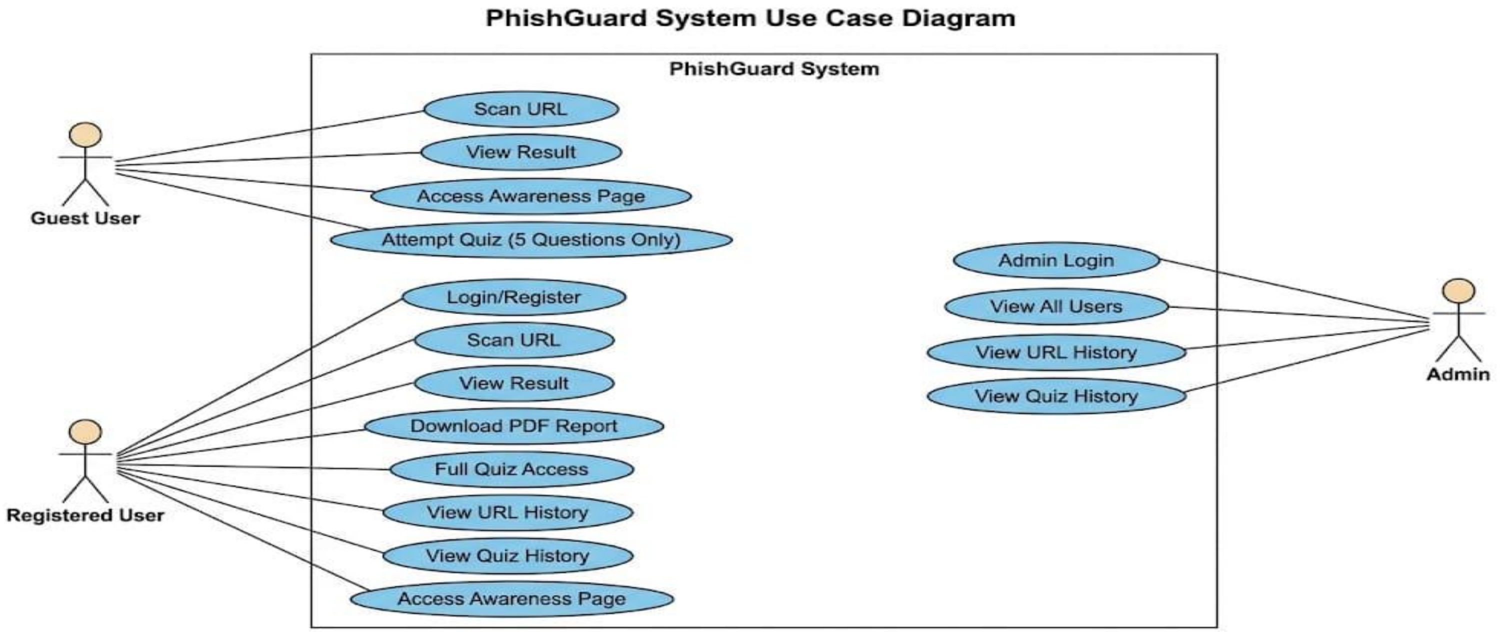


PhishGuard Component Diagram



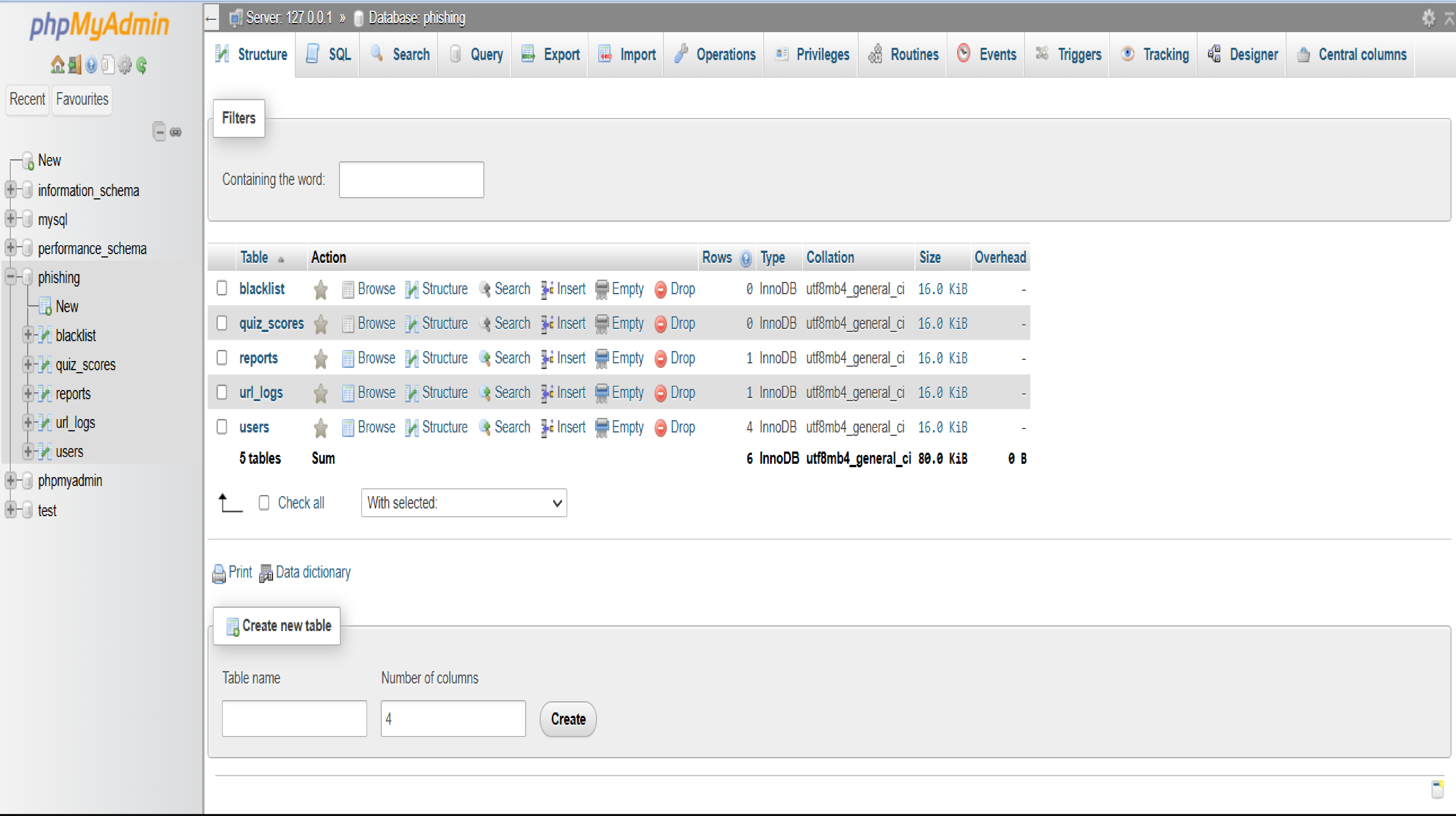
UML DIAGRAMS





DATABASE DESIGN

The database is designed to store all important system data such as user information, URL scan history, quiz results, and generated reports. This helps in maintaining proper records for each user. It includes tables like users, url_logs, quiz_scores, and reports, which manage different parts of the system efficiently.


phpMyAdmin
Recent
Favourites
New
information_schema
mysql
performance_schema
phishing
New
blacklist
quiz_scores
reports
url_logs
users
phpmyadmin
test
Server: 127.0.0.1 » Database: phishing
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
Routines
Events
Triggers
Tracking
Designer
Central columns
Filters
Containing the word:
Table
Action
Rows
Type
Collation
Size
Overhead
blacklist Browse Structure Search Insert Empty Drop 0 InnoDB utf8mb4_general_ci 16.0 KiB -
quiz_scores Browse Structure Search Insert Empty Drop 0 InnoDB utf8mb4_general_ci 16.0 KiB -
reports Browse Structure Search Insert Empty Drop 1 InnoDB utf8mb4_general_ci 16.0 KiB -
url_logs Browse Structure Search Insert Empty Drop 1 InnoDB utf8mb4_general_ci 16.0 KiB -
users Browse Structure Search Insert Empty Drop 4 InnoDB utf8mb4_general_ci 16.0 KiB -
5 tables Sum 6 InnoDB utf8mb4_general_ci 80.0 KiB 0 B
Check all
With selected:
Print
Data dictionary
Create new table
Table name
Number of columns
4
Create

OUTPUT SCREENS

The output screens show the actual working results of the system after user interaction. These screens demonstrate how the system responds to login, URL scanning, and quiz attempts. It includes login pages, URL result pages, dashboard view, and quiz result display, showing complete system functionality.

PhishGuard

Home Awareness Quiz Account

Result : SAFE

Risk Level

0% Risk Detected

Why this result?

- No suspicious patterns detected
- Uses HTTPS connection
- No blacklist match found

Scan Details

URL: https://in.pinterest.com/

Source: WhatsApp

IP: ::1

Time: 2026-05-18 21:42:41

Download Report

Login or signup required to download detailed report.

localhost/phishguard/quiz.php

PhishGuard

Home Awareness Quiz About Account

⚠ Guest Mode: Only 5 questions available. Login / Signup for full 20 questions and score history.

Your Score: 5 / 5

🔥 **Excellent Start!**

Q1. What is cybersecurity mainly meant to protect?

Your Answer: Internet-connected systems and data

Q2. Which of these is part of personal information?

Your Answer: All of these

Q3. Social engineering mainly attacks:

Your Answer: Human emotions and trust

ADVANTAGES

The system provides multiple benefits to users by making phishing detection simple and accessible. It helps users quickly identify unsafe websites and improves their understanding of cybersecurity threats. It also provides a user-friendly interface, fast detection, free access, and improved online safety awareness.

LIMITATIONS

Although the system is effective, it has some limitations due to its design. It uses rule-based detection and does not include advanced AI or machine learning techniques. It also has a limited blacklist database and may not detect highly advanced phishing attacks.

FUTURE SCOPE

The system can be improved in the future by integrating advanced technologies such as AI and machine learning. This will help in improving detection accuracy and identifying complex phishing attacks. Future improvements also include browser extensions, email phishing detection, and mobile application development for better accessibility.

CONCLUSION

PhishGuard is an effective system that helps users detect phishing websites and stay safe from cyber threats. It combines detection and awareness features in a simple and user-friendly platform. It improves cybersecurity knowledge and helps users browse the internet safely and confidently.

Thank You